

Netflix Streaming Log Analysis

Synthetic Dataset · 50,000 Sessions · Streaming QoS Intelligence

About the Dataset

This report is based on a synthetic dataset of 50,000 streaming session logs generated to simulate realistic user behaviour on a Netflix-like platform. The dataset captures three distinct user profiles — normal viewers, binge watchers, and sessions with poor network conditions — enabling meaningful outlier detection and clustering analysis.

Key Attributes

Attribute	Description
user_id / session_id	Anonymised user and session identifiers
timestamp	ISO 8601 datetime of the session
device	mobile / web / smart_tv / tablet
resolution	Stream quality: 480p / 720p / 1080p / 4K
buffering_ms	Time spent buffering (ms) — primary QoS metric
latency_ms	Network round-trip latency (ms)
watch_time_sec	Total seconds watched in the session
error	Error type if playback failed, else None
cdn	CDN serving the stream: Akamai / CloudFront / OpenConnect
region	Server region: us-east / us-west / eu-west / ap-south
status	Session outcome: success / failed
session_type	Ground-truth profile: normal / binge / bad

Problem Statement

Given real-time streaming session logs, automatically detect degraded user experiences caused by high buffering, network issues, or playback errors — and trigger intelligent corrective actions before the user drops off.

This pipeline answers five key questions:

- Which sessions are suffering from poor streaming quality?
- Is the problem caused by buffering, latency, or errors?
- Which CDN or region is underperforming?
- What automated action should the system take?
- Can we group sessions into experience tiers using unsupervised learning?

Dataset Composition

Session Type	Share	Count	Avg Buffering	Avg Latency
Normal	70%	35,000	~300 ms	~80 ms
Binge	20%	10,000	~500 ms	~100 ms
Bad	10%	5,000	~4,000 ms	~400 ms
Total	100%	50,000	—	—

Streaming Quality Analysis

The charts below show buffering distribution (with the 2,000 ms threshold marked), error frequency, average watch time by session type, and CDN performance. Bad sessions create a visible right-tail in the buffering histogram — these are the outliers driving the decision engine.

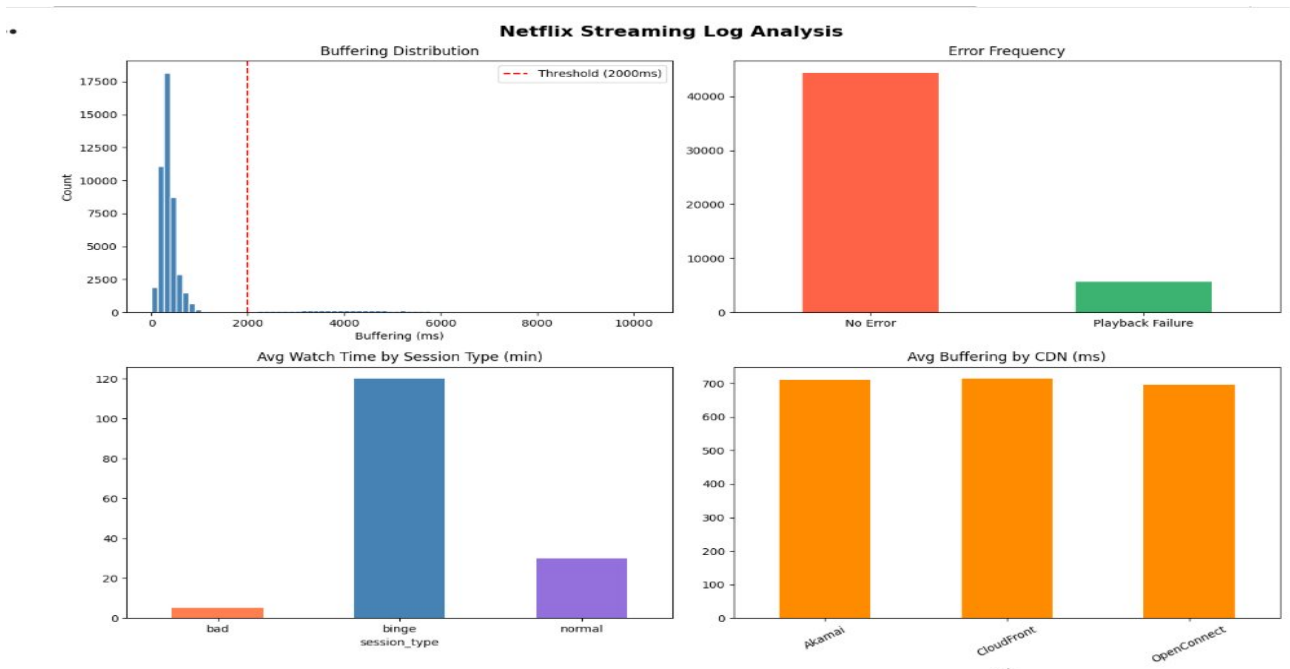


Figure 1 — Netflix Streaming Log Analysis: Buffering, Errors, Watch Time & CDN Performance

Dataset Distribution Overview

All nine attributes are plotted below. Notice that buffering and latency show right-skewed distributions due to the 10 % bad sessions, while device, resolution, CDN, and region are intentionally uniform (equal probability in generation).

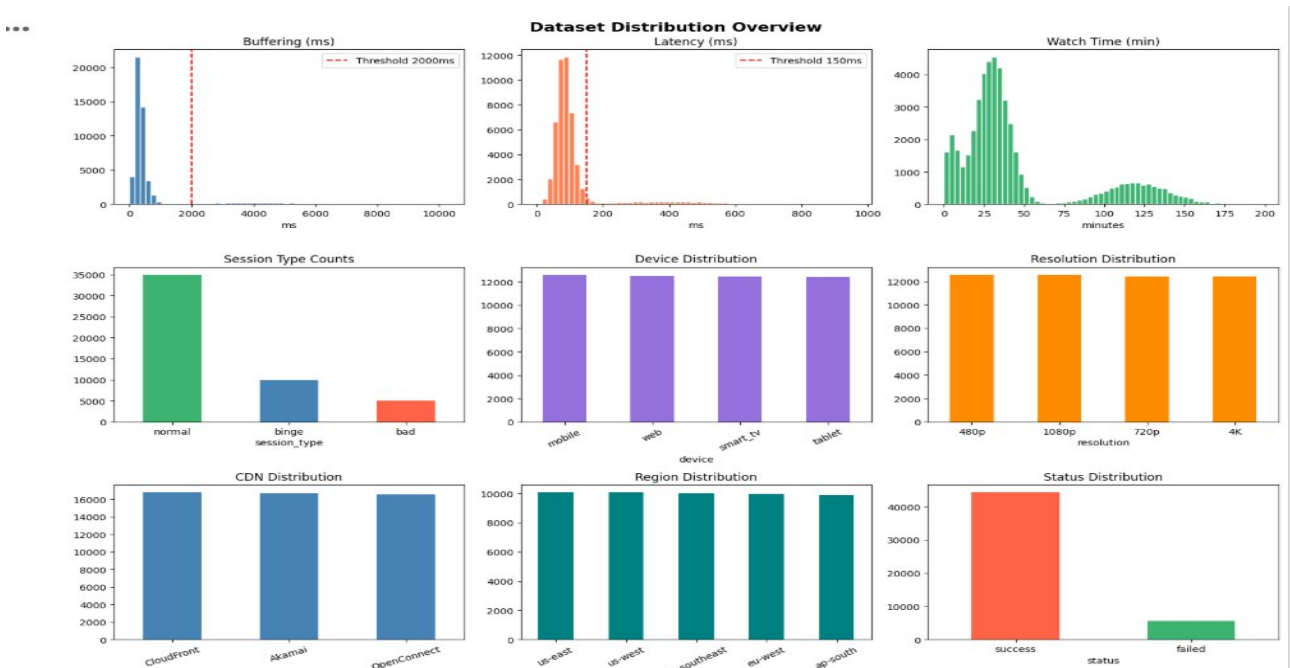


Figure 2 — Distribution of all key attributes across 50,000 sessions

Clustering & PCA Results

KMeans (k=3) was applied on buffering_ms, latency_ms, and watch_time_sec after StandardScaler normalisation. PCA reduced the three features to two components for visualisation. Three well-separated clusters emerged, each mapping to a distinct streaming experience tier.

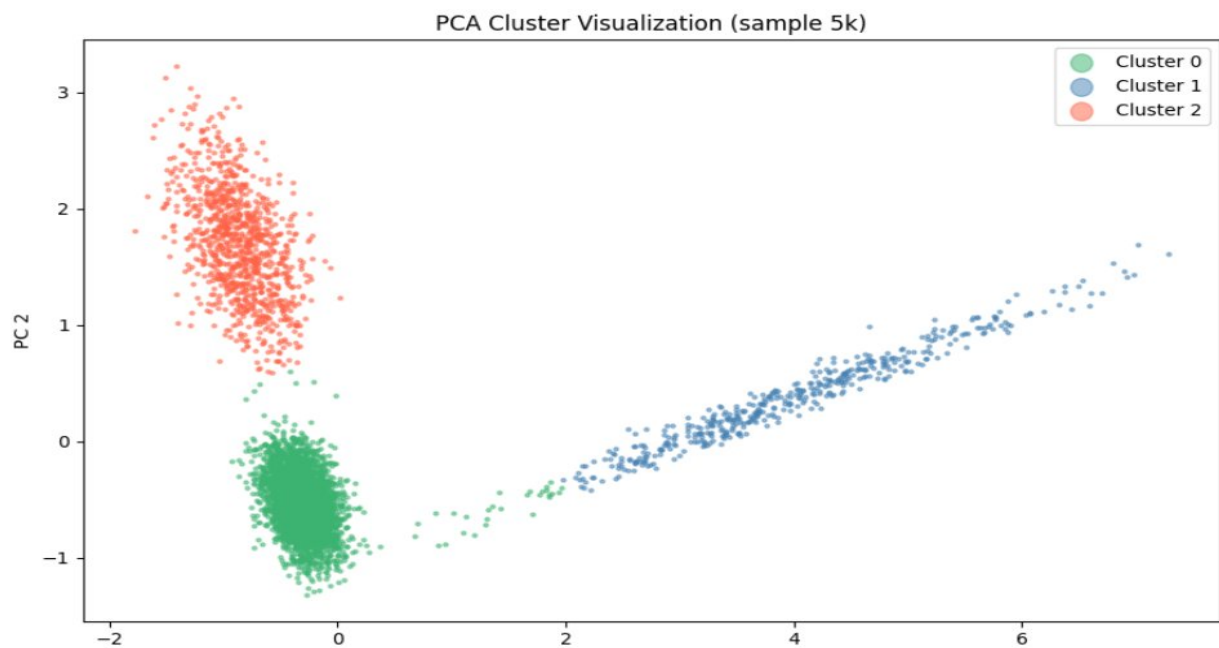


Figure 3 — PCA Cluster Visualisation (5,000-point sample)

Cluster Interpretation

The cluster summary table confirms the three tiers. Cluster 0 contains the bulk of sessions with smooth playback. Cluster 1 captures the bad sessions — high buffering (~4,097 ms), high latency (~409 ms), short watch time (~5 min), and a 55 % error rate. Cluster 2 represents binge sessions with long watch times and moderate buffering.

...	buffering_ms	latency_ms	watch_time_sec	watch_time_min	count	\
cluster						
0	314.3	80.4	1800.4	30.0	35425	
1	4097.4	409.6	308.5	5.1	4703	
2	500.6	99.5	7234.9	120.6	9872	
	error_rate_%					
cluster						
0	6.00					
1	55.01					
2	9.29					
Cluster Labels:						
	buffering_ms	latency_ms	watch_time_min	count		label
cluster						
0	314.3	80.4	30.0	35425	✓	Smooth Streaming
1	4097.4	409.6	5.1	4703	●	Severe Issues
2	500.6	99.5	120.6	9872	⚠	Moderate Issues

Figure 4 — Cluster summary: avg buffering, latency, watch time, count & error rate

Cluster Summary

Cluster	Label	Buffering	Latency	Watch Time	Count	Error Rate
0	Smooth Streaming	314 ms	80 ms	30 min	35,425	6.0 %
1	Severe Issues	4,097 ms	409 ms	5 min	4,703	55.0 %
2	Moderate Issues	500 ms	99 ms	120 min	9,872	9.3 %